[KEY] BOTANY



Trial Event

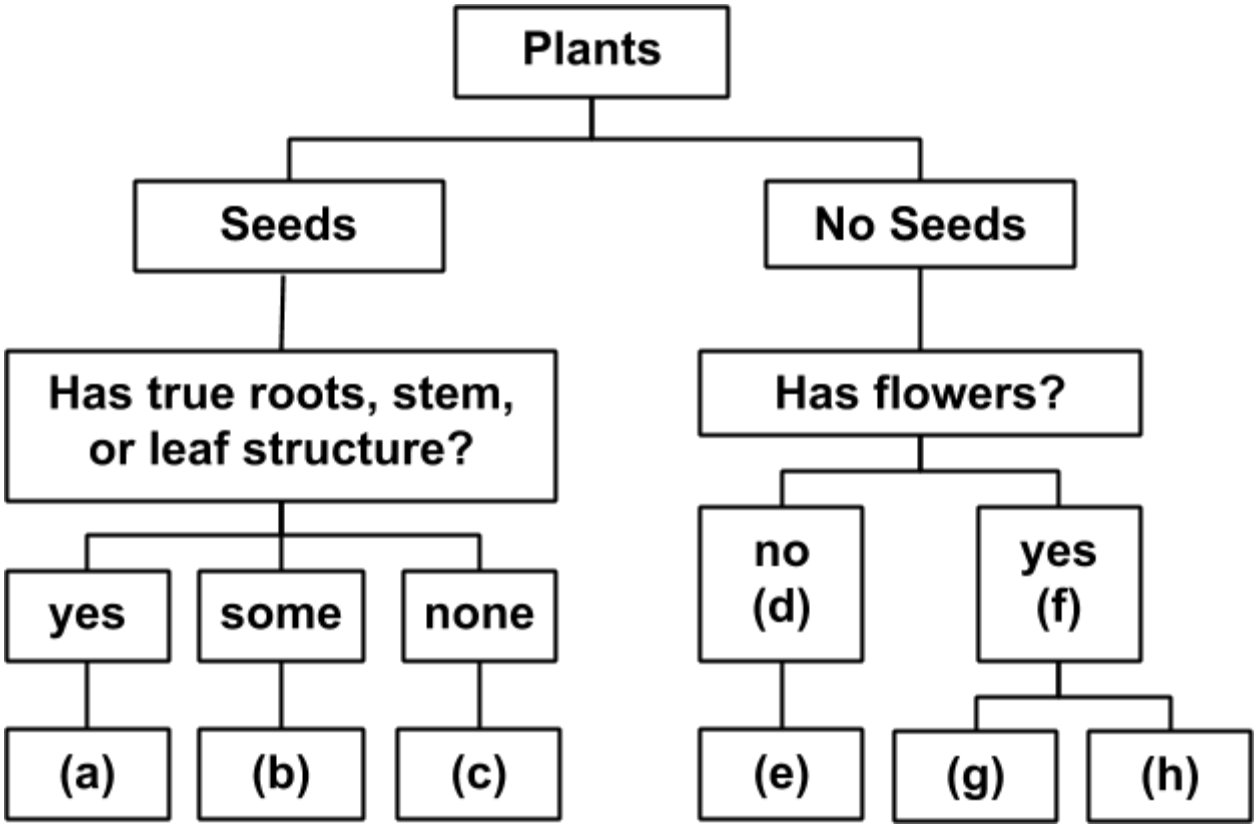
MIT Science Olympiad Invitational
January 25, 2020

Instructions:

Time: 50 Minutes
Number of Questions: 70
Number of Points: _____/200

Team Number:_____ Score:_____

Plant Classification (2 points per label)



a) Ferns

b) Mosses

c) Algae

d) Gymnosperms

e) Naked seeds/conifers

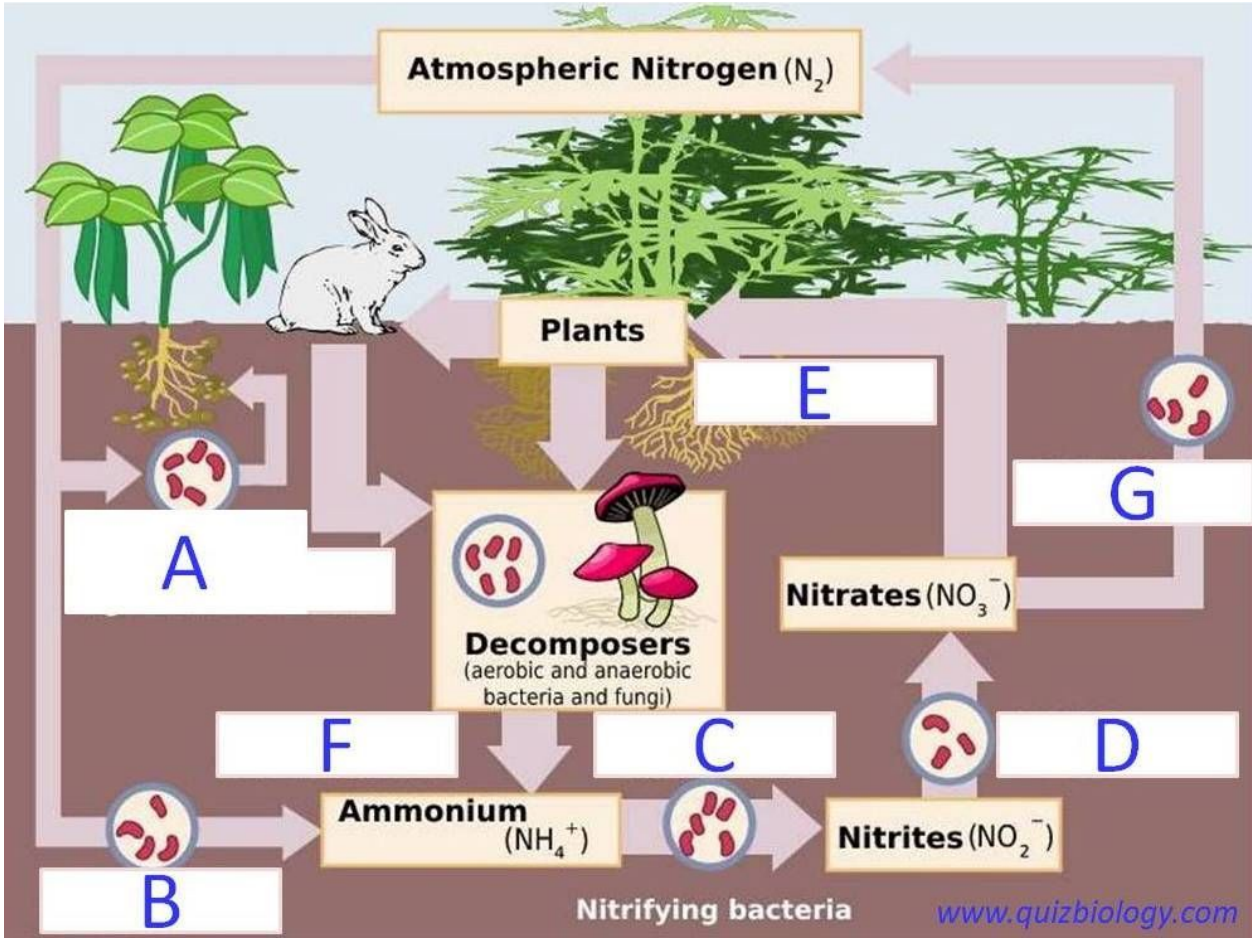
f) Angiosperms

g) Monocot/Dicot

h) Monocot/dicot

Team Number:_____ Score:_____

The Nitrogen Cycle (2 points per label)



a) Root nitrogen-fixing bacteria

b) Soil nitrogen-fixing bacteria

c) Nitrification

d) Nitrifying bacteria

e) Assimilation

f) Ammonification

g) Denitrifying bacteria

Team Number:_____ Score:_____

Multiple Choice (2 points each)

A____ 1. Starch is a polymer of

A. glucose B. sucrose C. fructose D. cellulose

B____ 2. The carbohydrate produced during the Calvin Cycle is

A. starch B. glucose C. PGA D. triose phosphate (PGald)

(PGA = phosphoglyceric acid; PGald = phosphoglyceraldehyde)

Use the following answer choices for questions 3-7.

A. glycolysis B. Krebs cycle C. electron transport chain (ETC) D. A and B

C____ 3. O2 (oxygen) is reduced to H2O (water)

B____ 4. pyruvic acid —>—>—> 3 CO2

C____ 5. a proton (H+) gradient forms and ATP is produced

A____ 6. glucose —>—>—> pyruvic acid

D____ 7. electrons are removed from NADH and FADH2

C____ 8. During photosynthesis, oxygen (O2) is released from

A. carbon dioxide (CO2) during the light reactions

B. carbon dioxide (CO2) during the Calvin cycle

C. water (H2O) during the light reactions

D. water (H2O) during the Calvin cycle

C____ 9. Ploidy refers to

A. homologous chromosomes B. the number of individual chromosomes in a nucleus

C. the number of sets of chromosomes in a nucleus

A____ 10. The phenomenon of double fertilization involves the fusion between

A. the egg and two sperm

B. the egg and one sperm and the polar nuclei (central cell) and the other sperm

C. the egg and one sperm and the synergids and the other sperm

D. the egg and one sperm and the antipodals and the other sperm

E____ 11. A spore

A. becomes a gametophyte by mitosis B. becomes a gametophyte by meiosis

C. becomes a sporophyte by mitosis D. becomes a sporophyte by meiosis E. is a gamete

D___ 12. A gametophyte produces

A. spores by mitosis B. spores by meiosis C. gametes by mitosis D. gametes by meiosis

B___ 13. A sporophyte produces

A. spores by mitosis B. spores by meiosis C. gametes by mitosis D. gametes by meiosis

Team Number:_____ Score:_____

D___ 14. The Calvin cycle of photosynthesis
A. occurs in the thylakoids and results in the production of ATP and NADPH
B. occurs in the stroma and results in the production of ATP and NADPH
C. occurs in the thylakoids and results in the production of carbohydrate (sugar)
D. occurs in the stroma and results in the production of carbohydrate (sugar)

B____ 15. The light reactions of photosynthesis
A. occur in the thylakoids and result in the production of ATP and NADPH
B. occur in the stroma and result in the production of ATP and NADPH
C. occur in the thylakoids and result in the production of carbohydrate (sugar)
D. occur in the stroma and result in the production of carbohydrate (sugar)

D____ 16. If a secondary cell wall is produced, it is often strengthened by the addition of
A. cellulose B. cutin C. calcium pectate (pectin) D. lignin

A____ 17. Plant cells that are usually dead at maturity include all of the following except
A. companion cells B. sclereids C. vessel elements D. tracheids

B____ 18. The tissue that moves dissolved sucrose in a generally downward direction is
A. xylem B. phloem C. tracheid D. vessel E. C and D

B____ 19. The structure that surrounds the cell and controls the movement of materials into and out of the cell is the
A. cell wall B. plasma membrane C. middle lamella D. plasmodesmata

A_____ 20. A plastid specialized for the storage of starch is the
A. amyloplast B. chloroplast C. chromoplast

D____ 21. The light colored, conducting xylem is called
A. hardwood B. summer wood C. heartwood D. sapwood

C____ 22. The tissue that divides the wood and the bark is the
A. cork B. periderm C. vascular cambium D. cork cambium E. secondary phloem

A____ 23. Lateral roots are produced by the
A. pericycle B. root hairs C. buds D. xylem E. endodermis

B____ 24. Wood is
A. secondary phloem B. secondary xylem C. A and B D. secondary xylem and bark

Team Number:_____ Score:_____

C____ 25. The dark, nonfunctional xylem at the center of a tree is called
A. hardwood B. late wood C. heartwood D. sapwood E. early wood

B____ 26. Growth rings can be seen in cross sections of trees because of differences in the sizes of the cells of the
A. sapwood and heartwood B. early wood and late wood
C. primary and secondary xylem D. primary and secondary phloem

C____ 27. Bark is a nontechnical name for
A. the cork B. the periderm C. all of the tissues outside the vascular cambium
D. all of the tissues outside the cork cambium

A____ 28. A nail is pounded into the side of a tree, 5 feet above the ground. The tree grows 1 foot per year. How far above the ground will the nail be after 3 years?
A. 5 feet B. 6 feet C. 7 feet D. 8 feet E. the tree will be dead by then

A____ 29. Loose collections of cells that allow gas exchange are called
A. lenticels B. stomata C. bud scales D. leaf scars

B____ 30. The roots of some plants are modified to parasitize other plants. These roots are called
A. mycorrhizae B. haustoria C. adventitious D. radicles

D____ 31. The poisonous nightshade plant is the ancestor of
A. Tomatoes B. Potatoes C. Chili and Bell Peppers D. All of the Above

A____ 32. Nitrifying bacteria ultimately creates
A. Nitrates B. Nitrites C. Ammonia D. Ammonium

B____ 33. Unlike most plants, these structures in cacti open at night, allowing them to thrive in hot and dry climates
A. lenticels B. stomata C. bud scales D. leaf scars

C____ 34. The Mendel pea plant experiment resulted in
A. genetic diversity of pea plants B. Punnett squares C. fundamental laws of inheritance

B____ 35. The plant type that produces the greatest percentage of atmospheric oxygen is
A. trees B. algae (phytoplankton) C. flowers D. succulents and cacti

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(answers may vary for the following)
3 points for correct answer, 2 points for any kind of extra explanation that makes sense

Propagating Produce (5 points each)

Questions 1-8.

You are standing in the produce section of your local supermarket, surveying the local selection. You want to start a small garden from kitchen scraps left over from cooking with the produce, rather than starting from seed. Indicate whether or not it would be possible to propagate a plant from a piece of produce for each plant type to produce a similar edible plant and reasoning why.

1. Carrots - no, carrots will regrow their greens from the top, but once the actual carrot part (taproot) is removed from the plant it won't regrow.
   Correct answer: no, Keywords: taproot, carrot greens, carrot top

2. Cavendish Banana - no, Cavendish bananas are propagated by pups or suckers, pieces of rhizome that form into miniature banana plants that can be severed from the parent and planted to become a separate plant.
   Correct answer: no, Keywords: pups/suckers, rhizome, split off plant/tree

3. Scallions/Green Onions (with roots) - yes, they are the top portions of onions, a root vegetable and with intact root structures will regrow in water
   Correct answer: yes, Keywords: onion, root vegetables, water propagation

4. Mango fruit - no, to bear fruit needs grafting
   Correct answer: no, Keywords: grafting, seeds sprout but infertile for fruit, produces a tree and leaves but not flowering

5. Avocado - no, to bear fruit needs grafting
   Correct answer: no, Keywords: grafting, seeds sprout but infertile for fruit, produces a tree and leaves but not flowering

6. Pineapple - yes, the top leaves are the beginning of the pineapple plant
   Correct answer: yes, Keywords: pineapple crown, leaves develop from inside outward, shrub development, flowering, berries

7. Lettuce/Cabbage - yes, lettuce will grow from its base
   Correct answer: yes, Keywords: head, leaves grow outwards, stump/bases, taproot

Team Number:_____ Score:_____

8. Lemon - no, while the seeds will sprout it also requires grafting
   Correct answer: no, Keywords: grafting, seeds sprout but infertile for fruit, produces a tree and leaves but not flowering

9. You notice a potato with green sprouts coming from its eyes. Is it still safe to eat or a potential poison? Use reasoning from the growth stages of tubers.
   No, at that point chlorophyll production has begun and solanine poisoning is a thing
   Correct answer: no, Keywords: sprouts, eyes, chlorophyll production, starch conversion, solanine poisoning, nightshade family relative

10. Oh no! Your lettuce/leafy green vegetable has wilted. Should you attempt to dry it in a salad spinner or soak it in water? Explain why for full credit.
    Soak in water, because osmosis and turgor pressure
    Correct answer: soak in water, Keywords: osmosis, turgor pressure, water pressure, cell rigidity, cell walls,

<u>Biodiversity (5 points each)</u>

1. Artificial banana flavoring does not taste like the Cavendish bananas we enjoy as fresh fruit today. Rather, it is the chemical isoamyl acetate associated with the flavor of the Gros Michel cultivar, now extinct due to fungal plagues such as Panama disease. Explain the importance of biodiversity in protecting our current banana crop for generations to come.
   If there's biodiversity in genes per plant they're less likely to all get wiped out by the same fungus (mutation gives that random probability)

2. Potatoes were once seen as poisonous and unfit for human consumption. Explain how human intervention led to the dispersion and adoption of the potato.
   Potato originated in South America, brought over through trade and colonization, peasants farmed and ate it as it produced numerously, selection of non-bitter tubers (glycoalkaloids)

3. Why is crop rotation important in the production of corn, soybeans, wheat, and/or sunflowers?
   so that the soil of farms is not used for only one set of nutrients. It helps in reducing soil erosion and increases soil fertility and yield crop, alternating deep and shallow roots

Team Number:_____ Score:_____

4. Genetically modified organisms (GMOs) are often a source of public scrutiny and skepticism. Provide examples of how GMO plants have positively and negatively affected agriculture.
   Ex: Monsanto, forcing farmers to rebuy seeds, vs solving global nutrition crisis with golden rice edited to have more vitamin A

5. Describe plant succession following a wildfire. What plants should be introduced on rocks and soil to revitalize a burned area?
   Keywords: lichens, annual plants, grasses, perennials

Hydroculture and Hydroponics (5 points each)

Tired of never knowing when it's right to water your plants (or overwatering them :( ) you decide to look into hydroculture and hydroponics, the process of growing plants without soil and with large amounts of water.

1. How is it possible for plants to be grown without soil? Explain using essential elements of plant growth and development.
   Keywords: rooting in alternative substrates, supplements for nitrogen in water, ammonia in water, a general nutrient solution

2. What types of nutrient supplements would be needed for hydroponic plants to thrive?
   Keywords: nitrogen, ammonia, liquid fertilizer, light waste, bacteria, potassium, phosphorus, micronutrients

3. Describe existing use of hydroculture in commercial farming (plants that are grown at scale while partially or entirely submerged in water).
   Ex. rice paddies, cranberry bogs, etc

4. Explain whether or not fish are beneficial for a hydroponic system.
   Fish are a source of nitrogen and ammonia, but don't expect them to get nutrients from the plant roots (NOT a symbiotic relationship and requires food be added)

5. Despite a typical hot and dry climate of origin, some types of succulents and cacti can be grown in water when certain conditions are met. What are these conditions?
   Cool and humidity at night, open stomata at night, wide shallow root networks to optimize rainfall, cells specialized to hold water.

MIT Invitational - Repaired practice edition - Answer key

Division C | Botany | Repair appendix

This appendix contains authored replacements and supplied written alternatives. It does not reconstruct unpublished historical material. For the listed item IDs, use this appendix instead of the archived wording and key.

ANSWER KEY - Review only after submitting.

repair-case. (4 points) Authored written substitute for unavailable source material: Distinguish hydroponics and aquaculture.

Unit: Horticulture, aquatic systems, medicine, and materials.
Worked-case data: Two irrigation treatments produce the same growth index, but one uses twice the water. What else is needed to choose a management approach?

answer: Hydroponics is a plant cultivation method using nutrient solution; aquaculture cultivates aquatic organisms and can include different systems. (4 points).